

Rajendra Bist

Full-Stack Web Developer & Backend Engineer

+977 9815625989 | rajendrabist396@gmail.com | linkedin.com/in/rajendra-bist | github.com/rajendrabist07 | rajendra-bist.vercel.app

SUMMARY

Full-stack engineer with a backend-first mindset, trained at Vcare Technical Institute and currently pursuing B.Tech in Computer Science at Indira Gandhi National Open University (IGNOU). Designs and ships AI-integrated, production-grade systems — from WebSocket event architecture to streaming LLM pipelines — with deliberate attention to API design, database schema modelling, and system reliability. Three deployed AI and real-time products in production. Seeking backend-heavy full-stack or AI engineering roles.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python (basic), HTML5, CSS3

Frontend: React 18, Next.js 15 (App Router), Tailwind CSS, Framer Motion, Responsive Design

Backend: Node.js, Express.js, REST API Design, WebSockets (Socket.io), JWT Authentication, Zod Validation

Databases: MongoDB Atlas, Mongoose ODM, Supabase (PostgreSQL), pgvector, Database Schema Design

AI / LLM: Gemini API, Groq API, Prompt Engineering, Streaming Interfaces, Upstash Redis, Vercel AI SDK

Tools & DevOps: Git, GitHub, Vercel, Railway, Clerk Auth, Docker (learning), Linux CLI, CI/CD

PROJECTS

EduMethod AI | Next.js, TypeScript, Groq API, Gemini API, Clerk, Supabase (pgvector), Upstash Redis | Live | GitHub

2026

- Architected a full-stack AI learning platform that converts any book photo or syllabus text into a structured 7-day study plan with spaced repetition scheduling, adaptive quizzes, and a conversational AI doubt-coach — end-to-end in a single user flow.
- Engineered a dual-LLM pipeline: Groq (Llama) for low-latency study plan generation and Gemini API for multimodal syllabus extraction from uploaded book images; custom prompt chains produce structured JSON learning paths consumed by the frontend.
- Production auth and persistence layer: Clerk for session management and protected routes; Supabase (PostgreSQL + pgvector) for user progress tracking, spaced repetition state, and vector-based content retrieval — Upstash Redis for caching API responses.

SocraticAI | Next.js 15, Node.js, Gemini API, MongoDB, Tailwind CSS | Live | GitHub

2026

- Built and deployed a chat-first AI assistant that enforces Socratic dialogue via custom system prompt engineering — responding to user questions with structured counter-questions that guide reasoning rather than delivering direct answers.
- Streaming architecture: Gemini API responses streamed token-by-token via Web API ReadableStream in a Next.js 15 serverless route; MongoDB Atlas logs sessions with TTL auto-expiry; in-memory rate limiting protects free-tier API quota in production.

WhatsApp Clone | React, Node.js, Express.js, Socket.io, Tailwind CSS | Live

2026

- Designed a bidirectional WebSocket event protocol (join-room, send-message, typing-status, disconnect) with Socket.io room isolation, handling concurrent multi-user sessions without message loss or event ordering violations.
- Decoupled frontend (Vercel) and backend (Railway) deployments with CORS-aware configuration, reflecting production multi-service architecture and independent deployment pipelines.

Developer Portfolio with Streaming AI Chat Agent | Next.js 15, Gemini API, MongoDB, Framer Motion | Live

2026

- Floating AI agent streams context-aware portfolio responses via Gemini 1.5 Flash and Web API ReadableStream; dark/light theme system, Framer Motion scroll-reveal animations, contact form with MongoDB persistence.

EDUCATION

B.Tech in Computer Science & Engineering (Ongoing)

Indira Gandhi National Open University (IGNOU)

Full-Stack Web Development

Vcare Technical Institute

Completed 2025

+2 Science

KMC — Kailali Multiple Campus

ADDITIONAL

Currently Building: Extending EduMethod AI with an adaptive quiz engine and spaced-repetition analytics dashboard

Backend Focus: System design patterns, JWT auth flows, microservice architecture, PostgreSQL query optimisation

Location: Nepal (UTC+5:45) — available for remote roles worldwide; open to onsite/hybrid in Nepal or India

Languages: Nepali (Native), Hindi (Fluent), English (Professional)